

Expected multi-utility representation $\Rightarrow$ Closed-graph continuity

Result — ai; produced by Claude Fable 5.1 (Cowork session).

0.1 Premises

- **Expected multi-utility representation.** There is a set U of functions $u : X \rightarrow \mathbb{R}$ such that for all $p, q \in \Delta(X)$: $p \succcurlyeq q$ if and only if $EU_u(p) \geq EU_u(q)$ for every $u \in U$.

0.2 Conclusion

- **Closed-graph continuity.** The set $\{(p, q) \in \Delta(X) \times \Delta(X) : p \succcurlyeq q\}$ is closed in the product topology.

0.3 Proof

$\{(p, q) : p \succcurlyeq q\} = \bigcap_{u \in U} \{(p, q) : EU_u(p) - EU_u(q) \geq 0\}$, an intersection of preimages of $[0, \infty)$ under continuous (indeed linear) maps on $\Delta(X) \times \Delta(X)$.

Record: topics/decision-theory/results/multi-utility-implies-closed-graph.yaml